Fiscal Year:	FY 2018	**Task Last Updated:**	FY 12/20/2017
PI Name:	Laughlin, Mitzi Ph.D.		
Project Title:	Assessment of the Time Courses of Leg Muscle Strength and Performance Losses During Prolonged Spaceflight using MARES		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mslaughl@Central.UH.EDU	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-743-0405
Organization Name:	University of Houston		
PI Address 1:	Health and Human Performance		
PI Address 2:	3855 Holman St., Garrison Rm. 105R		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77204-6015	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	02/23/2016	**End Date:**	08/24/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 8/24/2020 per HHC element/JSC and NSSC information (Ed., 12/10/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clarke, Mark SF Ph.D. (University of Houston) Layne, Charles S Ph.D. (University of Houston)		
Grant/Contract No.:	NNX16AF63G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Use the Muscle Atrophy Research and Exercise System (MARES) device to quantify the time course of changes in muscle strength and performance of crewmember volunteers during six-month International Space Station (ISS) missions. Establish baseline strength and performance capabilities of each participating crewmember from two pre-flight test sessions, track changes from that baseline during three in-flight test sessions, and track recovery to baseline levels during five post-flight test sessions. The project is currently ongoing and being coordinated by the European Space Agency (ESA) for flight. Several meetings have taken place to validate the ground and flight data collection procedures and another is scheduled in August at the European Astronaut Centre in Cologne. Continuation of Modulation of Muscle Function by Lower Limb Loading during Space Flight, grant NNX12AF04G.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed investigation seeks to systematically examine the relationships among muscle strength and neuro-motor control performance. By tracking these relationships over time during flight we will be able to identify any adaptive changes in integrated motor performance. If successful, this investigation should provide significant benefits to the Human Research Program by quantifying the aspects of the Program Requirements Document (PRD) Risk of Impaired Performance Due to Reduced Muscle Mass, Strength and Endurance embodied in IRP (Integrated Research Plan) Gap M24 (What is the time course of changes in muscle protein turnover, muscle mass and muscle function during long-duration spaceflight?) and M2 (What is the current status of in-flight/post-flight exercise performance capability?). Additionally, this investigation will provide valuable information to clinicians regarding muscle atrophy due to injury or disease on Earth.
Task Progress:	The grant NNX16AF63G, "Assessment of the Time Courses of Leg Muscle Strength and Performance Losses During Prolonged Spaceflight using MARES," began in February 2016 and is approaching the two-year anniversary. The study is designed to examine declines in muscular strength, function, and neuromotor control during long-duration spaceflight using the MARES device in the Columbus module on the ISS. Extensive progress has been made in the second year with data collection completed on one crewmember and three more crewmembers should complete the full protocol by February 2018. The SarcoLab-3 team, which includes this project, ESA SarcoLab project, and Russian scientists from IBMP (Institute of Biomedical Problems) met several times to develop a common processing and analysis protocol that met each of our research objectives. These meetings were facilitated by CADMOS and the Senar engineers. The data quality is excellent from the ISS although there have been many MARES related malfunctions during testing.
Bibliography Type:	Description: (Last Updated: 12/10/2020)
Abstracts for Journals and Proceedings	Laughlin M, Reed J, Layne C, Kozlovskaya I, Koryak Y, Albracht K, Staeudle B, Rittweger J. "SARCOLAB-3: Myotendinous and Neuromuscular Adaptation to Long-term Spaceflight." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017